WASP-50b

R-1836

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_150921` · created 2026-08-03T06:24:55+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457286.939 +/- 0.056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.05
Transit depth (Rp/Rs)^2	2.97 +/- 1.73 [%]
Orbital Inclination (inc)	85.71 +/- 3.0
Airmass coefficient 1 (a1)	2.49 +/- 1.41
Airmass coefficient 2 (a2)	-0.66 +/- 1.3
Scatter in the residuals of the lightcurve fit is	52.38 %
Best Comparison Star	#3 - [105, 456]
Optimal Aperture	1.11
Optimal Annulus	4.5
Transit Duration (day)	0.056 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.05 vs 0.1404 (deviation 0.63σ)
Duration fit vs published	0.056 d vs 0.0754 d (deviation 0.63σ)
Mid-time O–C	34.9 min
Depth SNR	3.4
Residual scatter	52.38 %

Light curve

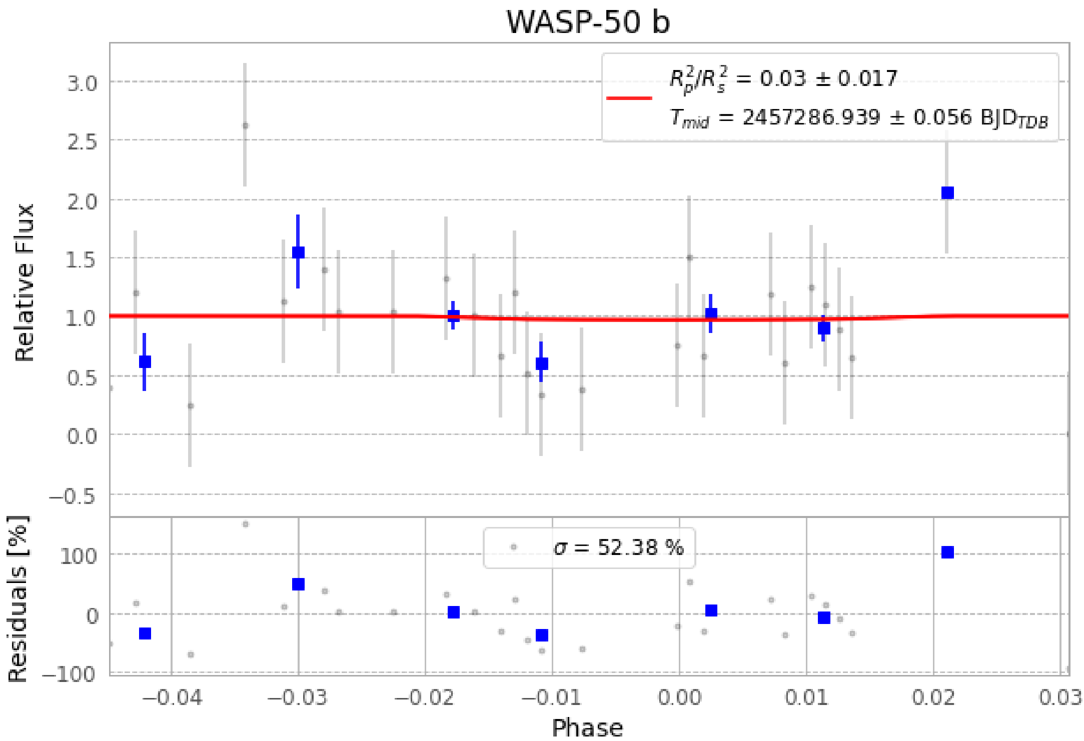

Figure 1 — FinalLightCurve_WASP-50 b_2015-09-21.png (SHA-256 7acfe3e03298eae5…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 425], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e2496b7f21e61982…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

74 FITS frames (49 MB), `WASP-50150921081214.FITS` → `WASP-50150921115112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-150921.log (2393f6f0abad…), FinalLightCurve_WASP-50 b_2015-09-21.png (7acfe3e03298…), FinalLightCurve_WASP-50 b_2015-09-21.csv (bf80efea7cc4…)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)